MATHEMATICS / INDEPENDENT STUDY

MM7030

Category Theory

Progressive hints

Original teaching material aligned with Stockholm University course descriptions.
Independently authored; not an official university publication.

Edition 2 · 25 October 2026

Contents

1 Unit 1 · Categories and duality 3
2 Unit 2 · Functors and equivalences 6
3 Unit 3 · Natural transformations 9
4 Unit 4 · Universal properties 12
5 Unit 5 · Limits and colimits 15
6 Unit 6 · Representability and Yoneda 18
7 Unit 7 · Adjunctions 21
8 Unit 8 · Preservation and applications 24
9 Mock assessment 1 27
10 Mock assessment 2 28

How to use this volume E problems practice definitions and standard arguments; C problems combine ideas; A problems require synthesis or generalization. These are study difficulty labels, not official SU grading criteria. Time estimates are guides, not deadlines. Write a complete attempt before opening hints. In the hints volume, read Hint 1 first and return to the problem before consulting Hint 2. Solutions give an argument and criteria for self-assessment. Expanded solutions also explain how to find the approach, a tempting dead end, and the role of the hypotheses.

Each unit has twelve core problems (four E, five C, three A) and an optional A-level transfer challenge numbered 13. The challenge asks you to choose a method in a new setting; budget its hour separately. Problem 12 and the challenge have detailed solutions with independently written decisive ideas. Mock estimates total 170 minutes of problem work plus ten minutes for review.

1 Unit 1 · Categories and duality

MM7030-U1.01 · E · 15 min
Learning objective: U1.1 Category axioms

Verify that nonnegative integers with an arrow $m \to n$ exactly when $m \leq n$ form a category.
Hint 1 — a starting point. Use reflexivity and transitivity.
Hint 2 — the key move. There is at most one arrow of each type.

MM7030-U1.02 · E · 15 min
Learning objective: U1.1 One-object categories

Describe the one-object category associated with the additive monoid $\mathbb{N}$. What arrows are invertible?
Hint 1 — a starting point. Composition is addition and identity is zero.
Hint 2 — the key move. An inverse would be a nonnegative integer adding to zero.

MM7030-U1.03 · E · 15 min
Learning objective: U1.2 Opposites

Describe the opposite of a poset category.
Hint 1 — a starting point. Reverse the order relation.
Hint 2 — the key move. An arrow $a \to b$ now means $b \leq a$.

MM7030-U1.04 · E · 15 min
Learning objective: U1.1 Isomorphisms

Prove an isomorphism in the category of groups is exactly a bijective group homomorphism.
Hint 1 — a starting point. The inverse of a bijection must also preserve the operation.
Hint 2 — the key move. Apply the original homomorphism to the desired inverse equation.

MM7030-U1.05 · C · 30 min
Learning objective: U1.2 Monomorphisms

Prove the composite of two monomorphisms is monic.
Hint 1 — a starting point. Cancel one factor at a time.
Hint 2 — the key move. From $nmu = nmv$, first cancel n.

MM7030-U1.06 · C · 30 min
Learning objective: U1.2 Cancellation

If gf is monic, prove f is monic. Must g be monic?
Hint 1 — a starting point. Precompose both sides of an equality with g.
Hint 2 — the key move. For the counterexample use an inclusion followed by a constant map on a larger set.

MM7030-U1.07 · C · 30 min
Learning objective: U1.2 Split arrows

Show a morphism with a left inverse is monic. Give a monomorphism of sets with no left inverse.
Hint 1 — a starting point. Use cancellation for the proof; consider the empty set for the example.
Hint 2 — the key move. The map from the empty set to a singleton is injective.

MM7030-U1.08 · C · 30 min
Learning objective: U1.2 Epimorphisms

Prove a function missing an element of its codomain is not epic in sets.
Hint 1 — a starting point. Construct two functions from the codomain to a two-element set.
Hint 2 — the key move. Make them differ only at the missing element.

MM7030-U1.09 · C · 30 min
Learning objective: U1.1 Groupoids

Show the isomorphisms in any category, retaining all objects, form a subcategory in which every arrow is invertible.
Hint 1 — a starting point. Check identities and composition.
Hint 2 — the key move. The inverse of gf is $f^{-1}g^{-1}$.

MM7030-U1.10 · A · 50 min
Learning objective: U1.2 Balancedness

Prove a morphism that is both monic and split epic is an isomorphism.
Hint 1 — a starting point. Let $fs = 1$.
Hint 2 — the key move. Compare $f(sf)$ with f.

MM7030-U1.11 · A · 50 min
Learning objective: U1.2 Rings versus sets

Prove the inclusion $\mathbb{Z} \to \mathbb{Z}\left[\frac{1}{2}\right]$ is epic among commutative unital rings but is not an isomorphism.
Hint 1 — a starting point. A map must send $\frac{1}{2}$ to the inverse of two.
Hint 2 — the key move. Inverses, when they exist, are unique.

MM7030-U1.12 · A · 50 min
Learning objective: U1.1 Categories from paths

Construct the free category on a directed graph and prove any assignment of its vertices and edges to objects and arrows of a category extends uniquely to a functor.
Hint 1 — a starting point. Use finite composable paths, including empty paths.
Hint 2 — the key move. A functor must send a path to the composite of its edge images.

MM7030-U1.13 · A · 60 min · Optional challenge

Learning objective: U1 Transfer: Words as arrows

Form a category with one object whose arrows are finite words in the two symbols a, b, including the empty word; composition is concatenation. Verify the category laws, determine which arrows are monomorphisms and epimorphisms, and determine which are isomorphisms.

Hint 1 — a starting point. Equality of words records every letter and its position.

Hint 2 — the key move. Concatenation has cancellation on either side; lengths add.

2 Unit 2 · Functors and equivalences

MM7030-U2.01 · E · 15 min
Learning objective: U2.1 Functor laws
Show a monotone map between posets gives a functor between their categories.
Hint 1 — a starting point. The object map is the given function.
Hint 2 — the key move. Monotonicity supplies the required arrow images.

MM7030-U2.02 · E · 15 min
Learning objective: U2.1 Forgetful functors
Define the forgetful functor from vector spaces over K to sets on objects and arrows.
Hint 1 — a starting point. Retain the underlying data.
Hint 2 — the key move. Linear maps are functions.

MM7030-U2.03 · E · 15 min
Learning objective: U2.1 Hom functors
For $f : X \to Y$, write the map $\mathcal{C}(A, f)$.
Hint 1 — a starting point. Postcompose an arrow from A.
Hint 2 — the key move. Its input is $h : A \to X$.

MM7030-U2.04 · E · 15 min
Learning objective: U2.2 Faithfulness
Explain why the forgetful functor from groups to sets is faithful.
Hint 1 — a starting point. Compare two homomorphisms with the same underlying function.
Hint 2 — the key move. Morphisms are actual functions with a property.

MM7030-U2.05 · C · 30 min
Learning objective: U2.1 Contravariance
Verify inverse image of subsets is a contravariant functor on sets.
Hint 1 — a starting point. Test identity and a composite.
Hint 2 — the key move. An element belongs to an inverse image exactly when its image belongs to the target subset.

MM7030-U2.06 · C · 30 min
Learning objective: U2.2 Fullness
Show the forgetful functor from groups to sets is not full.
Hint 1 — a starting point. Find a function which does not preserve the identity.
Hint 2 — the key move. Use a constant nonidentity value in a nontrivial group.

MM7030-U2.07 · C · 30 min
Learning objective: U2.2 Reflection of isomorphisms
Prove a full and faithful functor reflects isomorphisms.
Hint 1 — a starting point. Lift the inverse of Ff using fullness.
Hint 2 — the key move. Use faithfulness on the two inverse equations.

MM7030-U2.08 · C · 30 min
Learning objective: U2.1 One-object functors
Show functors between one-object categories of monoids are precisely monoid homomorphisms.
Hint 1 — a starting point. Only the arrow assignment is variable.
Hint 2 — the key move. Identity and composition become unit and multiplication.

MM7030-U2.09 · C · 30 min
Learning objective: U2.2 Matrix category
Show the category of matrices over K, with objects nonnegative integers, is equivalent to finite-dimensional vector spaces.
Hint 1 — a starting point. Map n to K^n.
Hint 2 — the key move. Every linear map between coordinate spaces has a unique matrix.

MM7030-U2.10 · A · 50 min
Learning objective: U2.2 Faithful reflection
Prove a faithful functor reflects monomorphisms and epimorphisms, without assuming fullness.
Hint 1 — a starting point. Apply the functor to the relevant equality.
Hint 2 — the key move. Cancel in the target, then use faithfulness.

MM7030-U2.11 · A · 50 min
Learning objective: U2.2 Equivalence of posets
Show equivalent poset categories are isomorphic as posets.
Hint 1 — a starting point. Natural isomorphisms between objects in a poset are equalities.
Hint 2 — the key move. A quasi-inverse is actually inverse on objects.

MM7030-U2.12 · A · 50 min
Learning objective: U2.2 Failure of preservation
Construct a faithful functor which sends a monomorphism and epimorphism to a function which is neither. Use the two-object poset $0 < 1$ as source, and explain why this does not contradict U2.10.
Hint 1 — a starting point. Every arrow in a poset category is monic and epic.
Hint 2 — the key move. Send the sole nonidentity arrow to a constant map between two-element sets.

MM7030-U2.13 · A · 60 min · Optional challenge

Learning objective: U2 Transfer: Discarding redundant points

Let objects be pairs (X, e) with X a finite set and $e : X \to X$ satisfying $e^2 = e$. An arrow $f : (X, e) \to (Y, d)$ satisfies $fe = f = df$, and the identity of (X, e) is e. Verify this is a category. Show that sending (X, e) to its image $e(X)$ and arrows to restrictions gives an equivalence with finite sets.

Hint 1 — a starting point. Each point of the image of an idempotent is fixed by that idempotent.

Hint 2 — the key move. Extend a function $h : e(X) \to d(Y)$ to X by $x \mapsto h(e(x))$.

3 Unit 3 · Natural transformations

MM7030-U3.01 · E · 15 min
Learning objective: U3.1 Naturality

Write the naturality equation for $\eta : F \Rightarrow G$ at $f : A \to B$.
Hint 1 — a starting point. Draw a square with FA, GA, FB, GB.
Hint 2 — the key move. Both paths must run from FA to GB.

MM7030-U3.02 · E · 15 min
Learning objective: U3.2 Identities

Show the identity maps 1_{FA} form a natural transformation $F \Rightarrow F$.
Hint 1 — a starting point. Insert them into the naturality equation.
Hint 2 — the key move. Both sides reduce to Ff.

MM7030-U3.03 · E · 15 min
Learning objective: U3.1 Natural projections

Show $\pi_1 : X \times X \to X$ is natural in the set X.
Hint 1 — a starting point. The product functor sends f to $f \times f$.
Hint 2 — the key move. Evaluate both composites on (x, y).

MM7030-U3.04 · E · 15 min
Learning objective: U3.2 Components

Describe a natural transformation between two constant functors with values A, B when the source is the terminal one-object category.
Hint 1 — a starting point. There is just one component.
Hint 2 — the key move. The only naturality equation involves identity arrows.

MM7030-U3.05 · C · 30 min
Learning objective: U3.1 Diagonal

Show the diagonal $\Delta_X : X \to X \times X$ is natural and each component is injective.
Hint 1 — a starting point. Evaluate on an element.
Hint 2 — the key move. Both square paths produce $(f(x), f(x))$.

MM7030-U3.06 · C · 30 min
Learning objective: U3.2 Vertical composition

Prove vertical composition of natural transformations is associative.
Hint 1 — a starting point. Compute the component at an arbitrary object.
Hint 2 — the key move. Use associativity in the target category.

MM7030-U3.07 · C · 30 min
Learning objective: U3.1 Inverse naturality

If $\eta : F \Rightarrow G$ is natural and each component is invertible, prove the inverse family is natural.
Hint 1 — a starting point. Rearrange the naturality equation by composing with inverses.
Hint 2 — the key move. Start from $Gf\eta_A = \eta_B Ff$.

MM7030-U3.08 · C · 30 min
Learning objective: U3.2 Group actions

For two actions of a group G on sets X, Y, show natural transformations of the corresponding functors are exactly equivariant functions.
Hint 1 — a starting point. There is one component, a function $T : X \to Y$.
Hint 2 — the key move. Naturality at g is the equivariance equation.

MM7030-U3.09 · C · 30 min
Learning objective: U3.1 Double dual

Verify naturality of the canonical map $V \to V^{**}$ by evaluating on a linear functional.
Hint 1 — a starting point. Equality of functionals can be checked at an arbitrary argument.
Hint 2 — the key move. For $f : V \to W$ and $\psi \in W^*$, both values are $\psi(f(v))$.

MM7030-U3.10 · A · 50 min
Learning objective: U3.1 Natural endomorphisms of sets

Classify all natural transformations from the identity functor on sets to itself without using Yoneda.
Hint 1 — a starting point. Use maps from a singleton selecting elements.
Hint 2 — the key move. The component on a singleton is forced to be its identity.

MM7030-U3.11 · A · 50 min
Learning objective: U3.2 Center of a group

Identify the natural endomorphisms of the identity functor of a group regarded as a one-object category.
Hint 1 — a starting point. A component is a group element.
Hint 2 — the key move. Naturality says it commutes with every group arrow.

MM7030-U3.12 · A · 50 min
Learning objective: U3.1 Scalar natural operations (proof reconstruction)

Show every natural endomorphism of the identity functor on K-vector spaces is multiplication by a single scalar.
Hint 1 — a starting point. Evaluate the component on the one-dimensional space at one.
Hint 2 — the key move. Use the linear map $K \to V$ taking one to a chosen vector.

MM7030-U3.13 · A · 60 min · Optional challenge

Learning objective: U3 Transfer: Uniform operations on subsets

Let P send a set to its power set and a function to direct image. Classify all natural transformations $P \Rightarrow P$ on the category of sets.

Hint 1 — a starting point. First determine what happens to the empty subset and to the full subset of a singleton.

Hint 2 — the key move. For a nonempty set A, use permutations of A, its map to a singleton, and its inclusion in a larger set.

4 Unit 4 · Universal properties

MM7030-U4.01 · E · 15 min
Learning objective: U4.1 Initial and terminal
Identify initial and terminal objects in sets and in groups.
Hint 1 — a starting point. Ask for unique maps in each direction.
Hint 2 — the key move. In groups, the trivial group serves both purposes.

MM7030-U4.02 · E · 15 min
Learning objective: U4.1 Products
Describe the product of a two-element set and a three-element set with its projections.
Hint 1 — a starting point. Use ordered pairs.
Hint 2 — the key move. The projections extract the two coordinates.

MM7030-U4.03 · E · 15 min
Learning objective: U4.2 Coproducts
Why is ordinary union not always a coproduct of two sets?
Hint 1 — a starting point. Consider two singleton sets with the same element.
Hint 2 — the key move. The two copies must be allowed to receive different images.

MM7030-U4.04 · E · 15 min
Learning objective: U4.1 Poset products
Find the product and coproduct of twelve and eighteen in positive integers ordered by divisibility.
Hint 1 — a starting point. Products are greatest lower bounds and coproducts least upper bounds.
Hint 2 — the key move. Use gcd and lcm.

MM7030-U4.05 · C · 30 min
Learning objective: U4.1 Product uniqueness
Prove two products of fixed objects are uniquely isomorphic compatibly with their projections.
Hint 1 — a starting point. Construct maps in both directions using universality.
Hint 2 — the key move. Their composites have the same projections as the identities.

MM7030-U4.06 · C · 30 min
Learning objective: U4.2 Finite biproducts
Prove $V \oplus W$ is both a product and a coproduct in vector spaces.
Hint 1 — a starting point. Use coordinate projections and inclusions.
Hint 2 — the key move. The mediators are pairing and addition of images.

MM7030-U4.07 · C · 30 min
Learning objective: U4.2 Free vector spaces
Construct the linear extension of a function from $\{a, b\}$ into a vector space, taking a to u and b to v.
Hint 1 — a starting point. Use linearity on the two basis vectors.
Hint 2 — the key move. A vector has unique coordinates $r[a] + s[b]$.

MM7030-U4.08 · C · 30 min
Learning objective: U4.1 Terminal products
Show $A \times 1$ is isomorphic to A whenever 1 is terminal and the product exists.
Hint 1 — a starting point. Pair the identity with the unique map to one.
Hint 2 — the key move. This pairing is inverse to the first projection.

MM7030-U4.09 · C · 30 min
Learning objective: U4.2 Groups and coproducts
Explain why $C_2 \times C_2$ with its coordinate inclusions is not a coproduct of two copies of C_2 in groups.
Hint 1 — a starting point. Map the two generators to noncommuting involutions in S_3.
Hint 2 — the key move. Use (12) and (23).

MM7030-U4.10 · A · 50 min
Learning objective: U4.1 Product associativity
Construct the canonical isomorphism $(A \times B) \times C \to A \times (B \times C)$ solely from universal properties.
Hint 1 — a starting point. Specify its three coordinate composites.
Hint 2 — the key move. Build nested pairings, then compare the coordinates of the two composites.

MM7030-U4.11 · A · 50 min
Learning objective: U4.2 Infinite sums and products
For countably many copies of K, show the finite-support direct sum is a coproduct but generally not the product under its coordinate projections.
Hint 1 — a starting point. Maps out of a direct sum use finite sums.
Hint 2 — the key move. A cone from K with every coordinate the identity would require an all-ones vector.

MM7030-U4.12 · A · 50 min
Learning objective: U4.1 Joint detection by projections
Prove product projections jointly determine every arrow into the product. Deduce the naturality of the canonical swap $A \times B \to B \times A$ in both variables.
Hint 1 — a starting point. Use uniqueness for the first claim.
Hint 2 — the key move. Both naturality-square composites have the same two projections.

MM7030-U4.13 · A · 60 min · Optional challenge

Learning objective: U4 Transfer: Joining pointed sets

In the category of pointed sets and basepoint-preserving functions, construct the coproduct of two finite pointed sets of sizes m, n. Prove its universal property, compute its size, and compare it with their product.

Hint 1 — a starting point. Start with a disjoint union, but consider what every basepoint-preserving map must do to its two distinguished points.

Hint 2 — the key move. Identify exactly the two basepoints, and no other elements.

5 Unit 5 · Limits and colimits

MM7030-U5.01 · E · 15 min
Learning objective: U5.1 Equalizers
Find the equalizer in sets of $f, g : \mathbb{Z} \to \mathbb{Z}$, $f(n) = n^2, g(n) = n$.
Hint 1 — a starting point. Solve the equality of the two values.
Hint 2 — the key move. The equation is $n(n-1) = 0$.

MM7030-U5.02 · E · 15 min
Learning objective: U5.1 Pullbacks
Find the pullback of $f : \mathbb{Z} \to \mathbb{Z}$, $f(a) = 2a$, and $g : \mathbb{Z} \to \mathbb{Z}$, $g(b) = 3b$, in sets.
Hint 1 — a starting point. Solve $2a = 3b$.
Hint 2 — the key move. Coprimality implies $a = 3k, b = 2k$.

MM7030-U5.03 · E · 15 min
Learning objective: U5.2 Coequalizers
In sets, coequalize the two maps from a singleton to $\{a, b, c\}$ selecting a and b.
Hint 1 — a starting point. Identify only what the maps force.
Hint 2 — the key move. The equivalence classes are $\{a, b\}$ and $\{c\}$.

MM7030-U5.04 · E · 15 min
Learning objective: U5.1 Empty diagrams
Explain why a limit of the empty diagram is terminal and a colimit is initial.
Hint 1 — a starting point. There is one empty family of maps for every apex.
Hint 2 — the key move. The mediator must therefore be unique for every object.

MM7030-U5.05 · C · 30 min
Learning objective: U5.1 Equalizers are monic
Prove an equalizer arrow is monic using only its universal property.
Hint 1 — a starting point. Consider two maps with the same composite through the equalizer.
Hint 2 — the key move. They are mediators of the same equalizing arrow.

MM7030-U5.06 · C · 30 min
Learning objective: U5.1 Pullbacks of monos
Prove a pullback of a monomorphism is monic.
Hint 1 — a starting point. Use the two projections to compare arrows into the pullback.
Hint 2 — the key move. Cancel the original monomorphism to obtain equality of the other projected composites.

MM7030-U5.07 · C · 30 min
Learning objective: U5.2 Linear coequalizers
Prove the coequalizer of linear maps $f, g : V \to W$ is $W/\operatorname{im}(f-g)$.
Hint 1 — a starting point. A linear map coequalizes exactly when it kills the difference.
Hint 2 — the key move. Use the universal property of a quotient vector space.

MM7030-U5.08 · C · 30 min
Learning objective: U5.2 Pushout of sets
Compute the pushout of inclusions $\{b\} \to \{a, b\}$ and $\{b\} \to \{b, c\}$ in sets.
Hint 1 — a starting point. Start with a tagged disjoint union.
Hint 2 — the key move. Identify the two copies of b.

MM7030-U5.09 · C · 30 min
Learning objective: U5.1 Limits in groups
Show the equalizer of group homomorphisms $f, g : G \to H$ is a subgroup of G. Must it be normal?
Hint 1 — a starting point. Test ab^{-1} for elements with equal images.
Hint 2 — the key move. For failure of normality compare an inner automorphism with the identity.

MM7030-U5.10 · A · 50 min
Learning objective: U5.1 Finite-limit construction
Construct the pullback of $f : A \to C, g : B \to C$ using a product and an equalizer.
Hint 1 — a starting point. Compare two arrows from $A \times B$ to C.
Hint 2 — the key move. Equalize $f\pi_A$ and $g\pi_B$.

MM7030-U5.11 · A · 50 min
Learning objective: U5.2 Gluing may identify more
Give a coequalizer in sets where the initially required identification pairs are not themselves transitive, and compute the actual quotient.
Hint 1 — a starting point. Use maps from a two-element set.
Hint 2 — the key move. Require $a \sim b$ and $b \sim c$.

MM7030-U5.12 · A · 50 min
Learning objective: U5.1 Pullback pasting
Prove that two horizontally composable pullback squares have a pullback as their outer rectangle.
Hint 1 — a starting point. Factor a compatible pair through the right square first.
Hint 2 — the key move. Then use the left square; reverse this reasoning for uniqueness.

MM7030-U5.13 · A · 60 min · Optional challenge

Learning objective: U5 Transfer: Compatible integer measurements

Fix a positive integer n. Find an abelian group P and maps $p, q : P \to \mathbb{Z}$ with p congruent to $2q$ modulo n, such that every pair $f, g : A \to \mathbb{Z}$ with this congruence factors uniquely through p, q. Give a basis for P.

Hint 1 — a starting point. Describe a pair of compatible integers by their difference $x - 2y$.

Hint 2 — the key move. Write $(x, y) = (nk + 2y, y)$.

6 Unit 6 · Representability and Yoneda

MM7030-U6.01 · E · 15 min
Learning objective: U6.1 Yoneda evaluation
For $\eta : \mathcal{C}(A, -) \Rightarrow F$, which element determines η under Yoneda?
Hint 1 — a starting point. Evaluate at the representing object.
Hint 2 — the key move. Use its identity arrow.

MM7030-U6.02 · E · 15 min
Learning objective: U6.2 Representing the identity
Show the identity functor on sets is covariantly represented by a singleton.
Hint 1 — a starting point. A function from a singleton selects an element.
Hint 2 — the key move. Evaluate at its unique point.

MM7030-U6.03 · E · 15 min
Learning objective: U6.1 Reconstructing a transformation
Let $x \in FA$. Define the corresponding transformation from $\mathcal{C}(A, -)$ to F.
Hint 1 — a starting point. Apply F to a map from A.
Hint 2 — the key move. The component at B sends f to $F(f)(x)$.

MM7030-U6.04 · E · 15 min
Learning objective: U6.2 Universal elements
For the underlying-set functor on groups represented by $\mathbb{Z}$, identify a universal element.
Hint 1 — a starting point. A map from $\mathbb{Z}$ is determined by one generator.
Hint 2 — the key move. Choose the integer one.

MM7030-U6.05 · C · 30 min
Learning objective: U6.1 Natural binary operations
Classify natural transformations $X \times X \to X$ in sets.
Hint 1 — a starting point. Represent pairs by maps from a two-element set.
Hint 2 — the key move. Yoneda gives one transformation for each element of that set.

MM7030-U6.06 · C · 30 min
Learning objective: U6.2 Contravariant representables
Given $f : A \to B$, write the natural transformation $\mathcal{C}(-, A) \Rightarrow \mathcal{C}(-, B)$ and prove naturality.
Hint 1 — a starting point. Postcompose by f.
Hint 2 — the key move. Precomposition with a test arrow commutes by associativity.

MM7030-U6.07 · C · 30 min
Learning objective: U6.2 Universal element category

Show a universal element of a covariant functor is an initial object in its category of elements.
Hint 1 — a starting point. Unpack arrows in the category of elements.
Hint 2 — the key move. Such an arrow is exactly a map satisfying $F(f)(u) = x$.

MM7030-U6.08 · C · 30 min
Learning objective: U6.1 Empty representing set

Classify natural transformations from the constant singleton functor on sets to the identity functor.
Hint 1 — a starting point. The constant singleton functor is represented by the empty set.
Hint 2 — the key move. Yoneda asks for an element of the empty set.

MM7030-U6.09 · C · 30 min
Learning objective: U6.2 Uniqueness

Prove that naturally isomorphic contravariant representables have isomorphic representing objects.
Hint 1 — a starting point. Use fullness of the Yoneda embedding on the isomorphism and its inverse.
Hint 2 — the key move. Faithfulness translates their composite identities back to arrows.

MM7030-U6.10 · A · 50 min
Learning objective: U6.1 Natural power operations on groups

Classify all natural transformations from the underlying-set functor on groups to itself.
Hint 1 — a starting point. It is represented by $\mathbb{Z}$.
Hint 2 — the key move. An integer n produces the operation $g \mapsto g^n$.

MM7030-U6.11 · A · 50 min
Learning objective: U6.2 Nonrepresentability

Show the constant empty-set functor from sets to sets is not covariantly representable.
Hint 1 — a starting point. Evaluate a supposed representation at its representing object.
Hint 2 — the key move. Every object has an identity arrow.

MM7030-U6.12 · A · 50 min
Learning objective: U6.2 Natural n-ary operations

For finite sets m, n, classify natural transformations $X^n \to X^m$ on sets, including the empty cases.
Hint 1 — a starting point. Use the covariant representables $\mathrm{Set}(n, -)$ and $\mathrm{Set}(m, -)$.
Hint 2 — the key move. Yoneda identifies transformations with functions $m \to n$.

MM7030-U6.13 · A · 60 min · Optional challenge

Learning objective: U6 Transfer: Families of subsets from pairs

For a function $f : X \to Y$, define $P(f)(S) = f[S]$ and $P(P(f))(\mathcal{A}) = \{f[S] : S \in \mathcal{A}\}$. Classify all families $\theta_X : X \times X \to P(P(X))$ satisfying $P(P(f))\left(\theta_{X(x,y)}\right) = \theta_{Y(f(x),f(y))}$ for every f. How many such families are there?

Hint 1 — a starting point. Study the ordered pair of distinct elements in a two-element set. Keep a family of subsets distinct from a single subset.

Hint 2 — the key move. The value at the generic pair is any collection of subsets of that two-element set. Transport the whole collection by direct image.

7 Unit 7 · Adjunctions

MM7030-U7.01 · E · 15 min
Learning objective: U7.1 Adjunction types

For $F : \mathcal{C} \to \mathcal{D}$ left adjoint to G, write the hom-set bijection and the types of the unit and counit components.

Hint 1 — a starting point. Place F on the source side of the target-category hom set.

Hint 2 — the key move. The unit starts at C, the counit ends at D.

MM7030-U7.02 · E · 15 min
Learning objective: U7.1 Currying

Curry the function $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$, $f(b, a) = 2b - a$.

Hint 1 — a starting point. Fix the first argument.

Hint 2 — the key move. The result is a function assigning a function.

MM7030-U7.03 · E · 15 min
Learning objective: U7.2 Evaluation

Describe the counit of the product–exponential adjunction in sets.

Hint 1 — a starting point. It evaluates a function at an argument.

Hint 2 — the key move. The domain is $X^A \times A$.

MM7030-U7.04 · E · 15 min
Learning objective: U7.2 Free-vector-space unit

Describe the unit of the free-vector-space/forgetful adjunction.

Hint 1 — a starting point. Send a generator to its formal basis vector.

Hint 2 — the key move. The source is a set, not a vector space.

MM7030-U7.05 · C · 30 min
Learning objective: U7.1 Power-set adjunction

Prove direct image along $f : X \to Y$ is left adjoint to inverse image on power-set posets.

Hint 1 — a starting point. Translate both inequalities into element membership.

Hint 2 — the key move. The condition is $f[S] \subseteq T$ iff $S \subseteq f^{-1}[T]$.

MM7030-U7.06 · C · 30 min
Learning objective: U7.1 Floor and inclusion

Let $i : \mathbb{Z} \to \mathbb{R}$ be inclusion as ordered sets. Show i is left adjoint to floor.

Hint 1 — a starting point. Compare an integer with a real number.

Hint 2 — the key move. For integer n, $n \leq x$ iff $n \leq \lfloor x \rfloor$.

MM7030-U7.07 · C · 30 min
Learning objective: U7.2 Free counit

For a vector space V, describe the counit $FUV \to V$ and verify the triangle involving the unit on UV.
Hint 1 — a starting point. A formal linear combination is evaluated as an actual sum.
Hint 2 — the key move. A single formal basis vector $[v]$ evaluates to v.

MM7030-U7.08 · C · 30 min
Learning objective: U7.2 Transpose formulas

Given unit and counit, write the transpose of $h : FC \to D$ and the inverse transpose of $k : C \to GD$.
Hint 1 — a starting point. Use one unit or one counit.
Hint 2 — the key move. The formulas are $Gh\eta_C$ and $\varepsilon_D Fk$.

MM7030-U7.09 · C · 30 min
Learning objective: U7.1 Diagonal adjunctions

Show the diagonal functor $\Delta : \mathcal{C} \to \mathcal{C} \times \mathcal{C}$ is left adjoint to the binary-product functor when products exist.
Hint 1 — a starting point. Maps from (X, X) to (A, B) are pairs of maps.
Hint 2 — the key move. The product universal property supplies the corresponding map into $A \times B$.

MM7030-U7.10 · A · 50 min
Learning objective: U7.2 Triangle converse

Prove the two unit–counit transpose formulas are inverse assuming naturality and the triangle identities.
Hint 1 — a starting point. Expand each round trip.
Hint 2 — the key move. Move the map past the counit or unit using naturality.

MM7030-U7.11 · A · 50 min
Learning objective: U7.1 Composite adjunctions

If $F \dashv G$ and $F' \dashv G'$ are composable adjunctions, prove $F'F \dashv \mathbb{G}'$.
Hint 1 — a starting point. Compose the two hom-set bijections.
Hint 2 — the key move. Track the order of the right adjoints.

MM7030-U7.12 · A · 50 min
Learning objective: U7.2 Closure operators

For an adjunction $F : P \to Q, G : Q \to P$ of posets, prove GF is monotone, extensive, and idempotent.
Hint 1 — a starting point. The unit says $p \leq GFp$.
Hint 2 — the key move. Use the counit inequality on Fp for the reverse idempotence inequality.

MM7030-U7.13 · A · 60 min · Optional challenge
Learning objective: U7 Transfer: A relation produces a closure

Given a relation $R \subset X \times Y$, let $E(A)$ be the set of points of Y related to some member of $A \subset X$. For $B \subset Y$, find the greatest subset $U(B) \subset X$ with $E(U(B)) \subset B$. Prove $A \subset U(E(A))$ and $U(E(U(E(A)))) = U(E(A))$.

Hint 1 — a starting point. Translate 'greatest' into a condition on each possible point x.

Hint 2 — the key move. Every point related to x must lie in B, including the case where x has no related points.

8 Unit 8 · Preservation and applications

MM7030-U8.01 · E · 15 min
Learning objective: U8.1 Preservation direction
Which of products and coproducts must a right adjoint preserve, when they exist?
Hint 1 — a starting point. Use the limit/colimit distinction.
Hint 2 — the key move. Products are limits.

MM7030-U8.02 · E · 15 min
Learning objective: U8.1 Terminal objects
Explain why a right adjoint preserves terminal objects.
Hint 1 — a starting point. Terminal objects are empty limits.
Hint 2 — the key move. Apply limit preservation.

MM7030-U8.03 · E · 15 min
Learning objective: U8.2 Abelianization
What is the abelianization of an abelian group?
Hint 1 — a starting point. All commutators are already the identity.
Hint 2 — the key move. The quotient is by the trivial subgroup.

MM7030-U8.04 · E · 15 min
Learning objective: U8.2 Exponentials
In sets, how many functions are in B^A if $|A| = 3$ and $|B| = 2$? Explain its role in currying.
Hint 1 — a starting point. Choose the image of each of three inputs.
Hint 2 — the key move. It is the object representing maps out of a product with A.

MM7030-U8.05 · C · 30 min
Learning objective: U8.1 Representable preservation
Show $\mathcal{C}(A, -)$ preserves a binary product.
Hint 1 — a starting point. Use the product property with source A.
Hint 2 — the key move. A map into the product is equivalent to a pair of component maps.

MM7030-U8.06 · C · 30 min
Learning objective: U8.1 Nonpreservation
Show the free-vector-space functor does not preserve the terminal set.
Hint 1 — a starting point. Compute the free vector space on one generator.
Hint 2 — the key move. The terminal vector space has dimension zero.

MM7030-U8.07 · C · 30 min
Learning objective: U8.2 Abelianization map
Compute the abelianization of S_3.
Hint 1 — a starting point. The commutator subgroup lies in the sign kernel.
Hint 2 — the key move. A commutator of two transpositions is a nontrivial three-cycle.

MM7030-U8.08 · C · 30 min
Learning objective: U8.2 Free coproducts
Construct the canonical isomorphism $F(A \coprod B) \cong F(A) \oplus F(B)$ for free vector spaces.
Hint 1 — a starting point. Send tagged basis vectors to the matching summand.
Hint 2 — the key move. Use linear extension in both directions.

MM7030-U8.09 · C · 30 min
Learning objective: U8.2 Implication in a poset
For subsets of S, prove $X \cap A \subseteq B$ iff $X \subseteq (S \setminus A) \cup B$.
Hint 1 — a starting point. Consider whether an element of X belongs to A.
Hint 2 — the key move. If it does, the first inclusion forces membership in B.

MM7030-U8.10 · A · 50 min
Learning objective: U8.1 Right-adjoint preservation
Prove directly through adjunct transposes that a right adjoint preserves an equalizer.
Hint 1 — a starting point. Transpose an arrow equalizing the image pair.
Hint 2 — the key move. Naturality transfers the equality to the original pair.

MM7030-U8.11 · A · 50 min
Learning objective: U8.2 Abelianization adjunction
Prove abelianization is left adjoint to inclusion of abelian groups, including naturality.
Hint 1 — a starting point. Every map to an abelian group kills commutators.
Hint 2 — the key move. Use uniqueness of quotient factorization under pre- and postcomposition.

MM7030-U8.12 · A · 50 min
Learning objective: U8.1 Equivalences preserve limits
Prove an equivalence of categories preserves all existing small limits.
Hint 1 — a starting point. Move a competing cone to an isomorphic object in the image.
Hint 2 — the key move. Use fullness to lift its arrows and faithfulness to lift the cone equations.

MM7030-U8.13 · A · 60 min · Optional challenge

Learning objective: U8 Transfer: A quotient that loses a kernel

On abelian groups define $Q(A) = A/2A$, with induced maps. Prove that Q preserves coproducts and coequalizers, but does not preserve all kernels.

Hint 1 — a starting point. Any map from A to a group in which twice every element is zero kills $2A$.

Hint 2 — the key move. For the failure, examine multiplication by two on $\mathbb{Z}$.

9 Mock assessment 1

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM7030-M1.01 · E · 15 min
Learning objective: U1 Cancellation

State the definition of epimorphism and prove every surjective set function is epic.
Hint 1 — a starting point. Compare two maps after precomposition.
Hint 2 — the key move. Evaluate at a preimage of each target element.

MM7030-M1.02 · E · 15 min
Learning objective: U4 Universal properties

State the coproduct property and identify the coproduct of two finite sets of sizes three and four.
Hint 1 — a starting point. Use tagged injections.
Hint 2 — the key move. The resulting set has seven elements.

MM7030-M1.03 · C · 30 min
Learning objective: U3 Naturality

Prove the swap maps $X \times X \to X \times X$, $(x, y) \mapsto (y, x)$ form a natural isomorphism.
Hint 1 — a starting point. Check a general function $f : X \to Y$.
Hint 2 — the key move. Both square paths send (x, y) to $(f(y), f(x))$.

MM7030-M1.04 · C · 30 min
Learning objective: U5 Limits

Construct the pullback in sets of two maps to a singleton and prove its universal property.
Hint 1 — a starting point. Compatibility is automatic.
Hint 2 — the key move. The construction is a Cartesian product.

MM7030-M1.05 · A · 40 min
Learning objective: U6 Yoneda

Classify natural transformations $X^3 \to X^2$ in sets and count them.
Hint 1 — a starting point. Represent the two powers by finite sets.
Hint 2 — the key move. They correspond to functions from a two-element set to a three-element set.

MM7030-M1.06 · A · 40 min
Learning objective: U7–U8 Adjunctions

Prove a left adjoint preserves initial objects directly from its hom-set bijection.
Hint 1 — a starting point. Count maps from the image of an initial object.
Hint 2 — the key move. The corresponding source hom set is a singleton.

10 Mock assessment 2

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM7030-M2.01 · E · 15 min
Learning objective: U2 Functors

Prove every functor preserves isomorphisms.
Hint 1 — a starting point. Apply it to both inverse equations.
Hint 2 — the key move. The image of an inverse is an inverse.

MM7030-M2.02 · E · 15 min
Learning objective: U7 Unit and counit

For free vector spaces, describe the unit at a set X and counit at a vector space V.
Hint 1 — a starting point. Distinguish generators from formal sums of actual vectors.
Hint 2 — the key move. Insertion and evaluation are the two maps.

MM7030-M2.03 · C · 30 min
Learning objective: U5 Coequalizers

Compute the coequalizer in vector spaces of $f, g : K \to K^2$ given by $f(t) = (t, 0)$ and $g(t) = (0, t)$.
Hint 1 — a starting point. Quotient by the image of $f - g$.
Hint 2 — the key move. The map $(a, b) \mapsto a + b$ has the required kernel.

MM7030-M2.04 · C · 30 min
Learning objective: U6 Representability

Show the functor $G \mapsto G \times G$ from groups to sets is represented by a free group on two generators.
Hint 1 — a starting point. A homomorphism from a free group is determined by generator images.
Hint 2 — the key move. Any ordered pair of group elements is permitted.

MM7030-M2.05 · A · 40 min
Learning objective: U2–U5 Equivalence

Let F be full, faithful, and essentially surjective. Prove it sends a terminal object to a terminal object.
Hint 1 — a starting point. Represent an arbitrary target-category object up to isomorphism by FA.
Hint 2 — the key move. Use the hom-set bijection supplied by full faithfulness.

MM7030-M2.06 · A · 40 min
Learning objective: U7–U8 Logic

In the power-set poset of S, identify a right adjoint to intersection with a fixed A and derive its counit inequality.
Hint 1 — a starting point. Solve $X \cap A \subseteq B$ for the largest permissible X.
Hint 2 — the key move. The largest choice is $(S \setminus A) \cup B$.